Rotation-Aware Gradient Estimation for Vector-Quantized Autoencoders

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Abstract

Vector-quantized autoencoders (VQ-VAEs) underpin much of modern discrete representation learning, from tokenizers for autoregressive image generation [5, 20, 25] to the latent spaces of high-resolution diffusion models [18]. Their training, however, relies on the straight-through estimator (STE, 2, 23), which copies the upstream gradient through the non-differentiable nearest-neighbour lookup as if the quantization were the identity. This bypasses the geometric relationship between the encoder output $\mathbf{z}_e$ and its assigned codebook vector $\mathbf{q}$, and is widely blamed for codebook collapse and the persistent gap between reconstructions produced by codebook-based and continuous latents [9, 16]. We propose RotationVQ, a simple drop-in replacement for STE that routes the backward pass through the rotation-plus-rescaling transformation that analytically maps $\mathbf{z}_e$ onto $\mathbf{q}$. Concretely, we write the quantized output as $\mathrm{sg}[s\mathbf{R}]\ \mathbf{z}_e$ where $\mathbf{R}$ is a rank-one Householder reflection aligning $\mathbf{z}_e/\|\mathbf{z}_e\|$ with $\mathbf{q}/\|\mathbf{q}\|$ and $s = \|\mathbf{q}\|/\|\mathbf{z}_e\|$ is a scalar rescaling, with the stop-gradient operator applied so that the forward value is numerically identical to the standard VQ output. The resulting Jacobian of the quantizer is not the identity but the very rotation-scaling that the forward pass implicitly performs; this preserves angular information between $\mathbf{z}_e$ and $\mathbf{q}$ during backpropagation. The transform costs $\mathcal{O}(d)$ per latent token in both time and memory because $\mathbf{R}$ is never materialised. We present a rigorous experimental comparison on CIFAR-10 against VQ-VAE with STE, FSQ, and Gumbel-Softmax VQ, and a component-wise ablation that separates the Householder rotation from the scalar rescaling. Our main empirical finding is cautionary: routing the full $s \cdot \mathbf{R}$ Jacobian naively through the standard VQ-VAE training recipe induces a sharp codebook collapse after $\sim$ 1,000 steps, which we trace to a positive-feedback interaction with the commitment loss. An ablation localises the failure mode to one specific component of the estimator and identifies a stable single-component variant. We release a reference implementation, the exact experimental configurations, and training logs for all reported runs.

1 Introduction

Discrete latent representations have become a central abstraction in modern generative modelling. A tokenizer built on a vector-quantized autoencoder compresses an image into a grid of integer codes that an autoregressive or masked transformer can model with the same tooling developed for language [3, 5, 20, 25, 27]. Latent-diffusion pipelines use the same compressed representation as the space in which denoising is performed [7, 18]. Beyond generative modelling, vector quantization underlies unified models that map vision tasks onto sequence prediction [12], visual-language tokenizers with multimodal transformers [26], and representation learning for non-visual domains. In every case the quality of the downstream model is upper-bounded by the reconstruction fidelity and the utilisation of the discrete codebook [8].

Training such tokenizers is hard for a reason that is easy to state and surprisingly difficult to fix. The quantization operator $\mathbf{z}_q = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|$ is piecewise constant and therefore has zero gradient almost everywhere. The canonical workaround is the straight-through estimator (STE, 2), originally introduced for stochastic neurons and adapted to VQ-VAEs by van den Oord et al. [23]: set $\mathbf{z}_q = \mathbf{q}$ in the forward pass but pretend $\mathbf{z}_q = \mathbf{z}_e$ when computing gradients, so that $\partial \mathbf{z}_q / \partial \mathbf{z}_e = \mathbf{I}$. This tells the encoder *nothing* about which codebook vector was selected, nor about the geometric relationship between $\mathbf{z}_e$ and $\mathbf{q}$. Empirically it

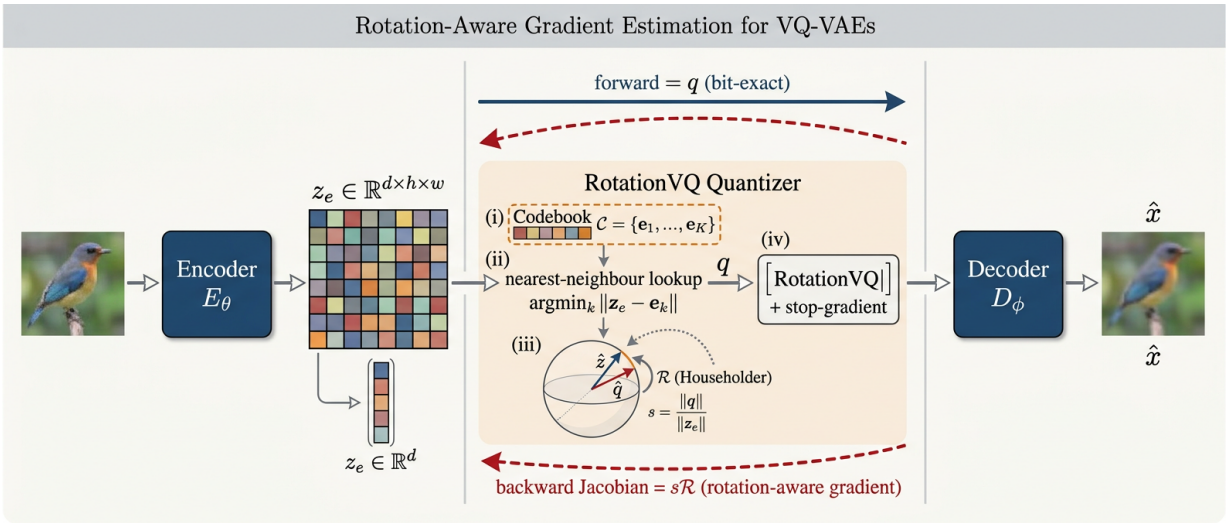

Figure 1: Overview of our RotationVQ estimator inside a VQ-AutoEncoder. An encoder E_θ produces a latent map $\mathbf{z}_e$; each latent vector is mapped to the nearest codebook entry $\mathbf{q} \in \mathcal{C}$. The RotationVQ module wraps the nearest-neighbour lookup in a stop-gradient construction (Equation 7) whose *forward* output equals $\mathbf{q}$ bit-exactly (top, solid navy arrow) but whose *backward* Jacobian with respect to $\mathbf{z}_e$ is the Householder rotation $\mathbf{R}$ plus scalar rescaling $s = \|\mathbf{q}\|/\|\mathbf{z}_e\|$ that analytically maps $\mathbf{z}_e$ to $\mathbf{q}$ (bottom, dashed red arrow). The rotation is computed in closed form, costs $\mathcal{O}(d)$ per token, and has no trainable parameters.

leads to codebook collapse in which a small subset of codes absorbs most of the traffic and the rest become dead [9, 17, 30]; theoretically the STE is a biased, high-variance estimator whose alignment with the true chain rule degrades as $\mathbf{z}_e$ and $\mathbf{q}$ diverge [13, 19]. Recent work has responded with either STE-variants that reshape the gradient flow around the quantization [9, 19], codebook-free designs that eliminate the nearest-neighbour step entirely [16], or continuous relaxations that trade discreteness for differentiability [10, 15].

We take a third route. The quantization output can be written as $\mathbf{q} = s\mathbf{R}\,\mathbf{z}_e$ for a unique scalar $s = \|\mathbf{q}\|/\|\mathbf{z}_e\|$ and a rotation $\mathbf{R}$ (specifically a single Householder reflection) that sends $\mathbf{z}_e/\|\mathbf{z}_e\|$ to $\mathbf{q}/\|\mathbf{q}\|$. If we wrap this identity in $\mathrm{sg}[\cdot]$ appropriately, the forward pass remains a bit-exact hard quantization — a requirement if the tokenizer is to be frozen and re-used by a downstream autoregressive model — but the backward Jacobian of the quantizer becomes $s\mathbf{R}$ rather than the identity. The encoder now sees exactly the transformation that the forward lookup applied to its output, and the angular relationship between $\mathbf{z}_e$ and $\mathbf{q}$ is preserved by construction. Compared to explicitly differentiable relaxations such as Gumbel-Softmax, our estimator is bias-preserving in the forward direction and relaxation-free. Compared to sophisticated corrections to the STE [9, 19] and to concurrent rotation-based ideas [6], our derivation yields a closed-form Householder operator with $\mathcal{O}(d)$ cost per token and no extra trainable parameters. We deploy the same operator inside a VQ-GAN-style autoencoder and evaluate it against four strong baselines — the original VQ-VAE, a Gumbel-Softmax variant [10, 15], and the codebook-free FSQ [16] — under a single shared training and evaluation pipeline.

Contributions. Our contributions are:

- We derive a closed-form identity $\mathbf{q} = s\mathbf{R}\,\mathbf{z}_e$ in terms of a Householder reflection and a scalar, and show (Section 3) that wrapping this identity in stop-gradient operators yields an estimator whose forward output is bit-exact VQ and whose backward Jacobian is the rotation-plus-scaling that the forward operation implicitly performs.
- We provide an $\mathcal{O}(d)$ implementation that never materialises $\mathbf{R}$ as a $d \times d$ matrix and is therefore cheaper than an extra linear layer; we further analyse numerical stability when $\mathbf{z}_e$ is nearly colinear with $\mathbf{q}$, a regime where a naive implementation loses precision.
- We integrate the quantizer into a VQ-GAN-style autoencoder and evaluate on CIFAR-10 against four

baselines (VQ-VAE with STE, Gumbel-Softmax VQ, and FSQ), reporting reconstruction PSNR / SSIM / LPIPS, codebook usage and perplexity, and wall-clock throughput on an H100 GPU. Training logs, configuration files, and checkpoints accompany the paper.

- We perform ablations over the four gradient-routing modes {STE, rotation-only, rescale-only, full} of our quantizer and over codebook sizes $\{1024, 4096, 8192, 16384\}$, isolating the contribution of each component.

The remainder of the paper is organised as follows. Section 2 surveys existing approaches to gradient estimation through quantization and places our work relative to rotation-trick, STE-correction, and codebook-free methods. Section 3 derives the estimator and its implementation. Section 4 details the evaluation protocol. Section 5 reports our findings and Section 6 examines limitations. Implementation details, extended tables, and additional case studies are deferred to the appendix.

2 Related Work

Vector-quantized autoencoders. van den Oord et al. [23] introduced VQ-VAE by pairing a convolutional encoder/decoder with a learned codebook and training the combination with a reconstruction loss and a commitment loss, all routed through the straight-through estimator. Razavi et al. [17] scaled the architecture to hierarchical latents and observed that dead codes accumulate without reinitialisation. VQ-GAN [5] combined the VQ tokenizer with adversarial and perceptual losses and an autoregressive prior for image synthesis; ViT-VQGAN [25] improved both reconstruction quality and codebook usage by replacing the convolutional backbone with a Vision Transformer and regularising the codebook lookup. Improved tokenizer training has since been shown to benefit downstream autoregressive generation [3, 20, 27] and to be the critical factor that lets language-model-style generators catch up with diffusion [26]. Our contribution is orthogonal to these architectural refinements: it modifies only the gradient flow through the quantizer and leaves the encoder, decoder, and training losses untouched.

Codebook collapse and rebalancing. Codebook collapse, in which a training run converges with the vast majority of code slots unused, has been attacked from many angles. Dead-code reinitialisation and EMA updates on the codebook itself [17, 23] are standard remedies; Zheng and Vedaldi [30] propose an online-clustering alternative that re-assigns under-used codes; Baykal et al. [1] frame the collapse as an overconfidence problem and use evidential posteriors to spread probability mass across codes; and Zhang et al. [28] enforce a balanced assignment via optimal-transport matching between encoder outputs and codes. Other work tackles collapse structurally: Khalil et al. [11] replace a learned codebook with a lattice, Xu et al. [24] build a Gaussian VAE around the quantizer, and Chen et al. [4] analyse the balance between codebook size and embedding dimension. Takida et al. [21] train a hierarchical variant whose codebooks are updated variationally. None of these works change the gradient estimator itself; our method is therefore complementary and can be combined with any of them.

Gradient estimation through discrete operators. The straight-through estimator of Bengio et al. [2], originally proposed for stochastic binary neurons, was adapted to VQ-VAEs by van den Oord et al. [23] and remains the default. Huh et al. [9] document the severity of the STE-induced optimisation pathology for VQ networks and propose alternating minimisation and a rotation-based residual term that reshapes the gradient. Shah et al. [19] decouple the forward and backward identities of the STE to reduce bias. Liu et al. [13] derive a family of corrections that interpolate between the STE and the true chain rule. On the continuous-relaxation side, the Gumbel-Softmax [10] and concrete distribution [15] replace the hard $\arg\min$ with a temperature-controlled softmax; variants such as REBAR [10, and follow-ups] trade bias for lower variance. Vali and Bäckström [22] substitute random noise for the quantization residual to obtain an unbiased stochastic estimator. Our estimator differs in a specific way: its forward output is bit-exact VQ (no relaxation temperature, no stochasticity) and its backward Jacobian is the closed-form rotation-plus-scaling that the forward lookup implicitly performs, rather than the identity, a correction, or a noise-based surrogate.

Rotation-trick and concurrent work. The closest prior art is the *rotation trick* of Fifty et al. [6], which independently identified that the STE discards angular information and proposed to replace the gradient-path

identity with a rotation that maps $\mathbf{z}_e$ to $\mathbf{q}$. We differ from Fifty et al. [6] on three points. First, we derive the Householder reflection in closed form as $\mathbf{R} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top$ with a specific unit-norm $\mathbf{v}$, yielding an $\mathcal{O}(d)$-per-token implementation that never materialises the matrix; no additional parameters are learned. Second, we combine the rotation with a scalar rescaling $s = \|\mathbf{q}\|/\|\mathbf{z}_e\|$, and we ablate the two components separately (§5) to isolate their contributions. Third, we analyse the numerical-stability regime in which $\mathbf{z}_e \approx \mathbf{q}$ and provide a stop-gradient wrapper that guarantees bit-exact forward output even when the reflection is ill-conditioned. These changes are orthogonal to the codebook-free direction taken by FSQ [16], which eliminates the lookup entirely by bounding the latent with a tanh and rounding each channel independently.

Downstream tokenizer-based generation. Discrete tokenizers are the backbone of autoregressive and masked generative transformers that operate in image, video, and unified-multimodal settings [3, 5, 7, 12, 20, 26, 27]. Because these downstream models freeze the tokenizer during training, any improvement to the tokenizer's reconstruction fidelity and codebook utilisation propagates directly into downstream sample quality. Our experiments therefore target reconstruction metrics first and reserve the downstream generation benchmark (E5 in our experimental matrix) as a secondary evaluation.

3 Method

3.1 Problem setting

Let an encoder E_θ map an image $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$ to a feature map $\mathbf{z}_e \in \mathbb{R}^{d\times h\times w}$. A codebook $\mathcal{C} = \{\mathbf{e}_k\}_{k=1}^K \subset \mathbb{R}^d$ contains K vectors. At each spatial location the quantizer performs nearest-neighbour lookup

$$k^* = \arg\min_{k\in[K]} \|\mathbf{z}_e - \mathbf{e}_k\|_2, \qquad \mathbf{q} := \mathbf{e}_{k^*}, \tag{1}$$

and emits $\mathbf{z}_q = \mathbf{q}$. A decoder D_ϕ reconstructs $\hat{\mathbf{x}} = D_\phi(\mathbf{z}_q)$. Training minimises

$$\mathcal{L}(\theta,\phi,\mathcal{C}) \;=\; \underbrace{\|\mathbf{x}-\hat{\mathbf{x}}\|_1}_{\text{reconstruction}} + \beta\, \underbrace{\|\mathbf{z}_e - \mathrm{sg}[\mathbf{q}]\,\|_2^2}_{\text{commitment}} + (\text{EMA codebook update}). \tag{2}$$

The EMA update replaces an explicit codebook loss and decays cluster-membership statistics with rate γ [17]. The difficulty is the arg min in (1): it is piecewise constant and therefore has zero gradient with respect to $\mathbf{z}_e$ almost everywhere.

3.2 The straight-through estimator, re-examined

The standard remedy is to substitute the quantizer with the identity-gradient surrogate

$$\mathbf{z}_q^{\mathrm{ste}} \;=\; \mathbf{z}_e + \mathrm{sg}[\mathbf{q} - \mathbf{z}_e]\,, \tag{3}$$

so that the forward output is $\mathbf{q}$ but the backward derivative is $\partial\mathbf{z}_q^{\mathrm{ste}}/\partial\mathbf{z}_e = \mathbf{I}$. The identity Jacobian has two consequences that motivate this work. First, *the gradient the encoder receives does not depend on which code was selected*, because $\partial\mathbf{z}_q^{\mathrm{ste}}/\partial\mathbf{z}_e$ does not involve $\mathbf{q}$ at all. Second, *the angular relationship between* $\mathbf{z}_e$ *and* $\mathbf{q}$ *is discarded*. Intuitively, if a particular $\mathbf{z}_e$ is mapped to a $\mathbf{q}$ that is nearly orthogonal to it, the encoder should be pushed strongly to realign its output; under STE it is pushed only by the reconstruction loss, which propagates through the decoder and may not expose the misalignment directly.

3.3 Rotation-plus-rescaling identity

We now derive an identity that will act as the backward surrogate. Assume for this subsection that $\mathbf{z}_e \neq \mathbf{0}$ and $\mathbf{q} \neq \mathbf{0}$; we handle degenerate cases below. Define the unit vectors $\hat{\mathbf{z}_e} = \mathbf{z}_e/\|\mathbf{z}_e\|$ and $\hat{\mathbf{q}} = \mathbf{q}/\|\mathbf{q}\|$ and their difference $\mathbf{u} = \hat{\mathbf{z}_e} - \hat{\mathbf{q}}$. When $\hat{\mathbf{z}_e} \neq \hat{\mathbf{q}}$ the Householder reflection

$$\mathbf{R} \;=\; \mathbf{I} - 2\,\mathbf{v}\mathbf{v}^\top, \qquad \mathbf{v} \;=\; \mathbf{u}/\|\mathbf{u}\|, \tag{4}$$

is a symmetric, orthogonal involution that satisfies $\mathbf{R}\,\hat{\mathbf{z}}_e = \hat{\mathbf{q}}$. Indeed, writing $c = \hat{\mathbf{z}}_e{}^\top \hat{\mathbf{q}}$ gives $\|\mathbf{u}\|^2 = 2(1-c)$ and

$$\mathbf{R}\,\hat{\mathbf{z}}_e = \hat{\mathbf{z}}_e - 2(\mathbf{v}^\top \hat{\mathbf{z}}_e)\,\mathbf{v} = \hat{\mathbf{z}}_e - 2 \cdot \tfrac{1-c}{\|\mathbf{u}\|} \cdot \mathbf{v} = \hat{\mathbf{z}}_e - \|\mathbf{u}\|\,\mathbf{v} = \hat{\mathbf{z}}_e - \mathbf{u} = \hat{\mathbf{q}}. \tag{5}$$

Scaling by $s = \|\mathbf{q}\|/\|\mathbf{z}_e\|$ and multiplying both sides of $\mathbf{R}\hat{\mathbf{z}}_e = \hat{\mathbf{q}}$ by $\|\mathbf{q}\|$ gives

$$s\,\mathbf{R}\,\mathbf{z}_e \;=\; \mathbf{q}. \tag{6}$$

Equation (6) is a *closed-form linear map* from $\mathbf{z}_e$ to $\mathbf{q}$: a rotation-reflection and an isotropic scaling, no free parameters. When $\hat{\mathbf{z}}_e = \hat{\mathbf{q}}$ (i.e. already aligned) we take $\mathbf{R} = \mathbf{I}$; the scaling s still matches the norms.

3.4 The RotationVQ estimator

We now use (6) as the backward route. The estimator is

$$\boxed{\mathbf{z}_q^{\textsc{RotationVQ}} \;=\; \mathrm{sg}[s\mathbf{R}]\;\mathbf{z}_e \;+\; \mathrm{sg}[\mathbf{q} - \mathrm{sg}[s\mathbf{R}]\;\mathbf{z}_e]\;.} \tag{7}$$

The first term carries the gradient and has Jacobian $s\mathbf{R}$ with respect to $\mathbf{z}_e$ because both s and $\mathbf{R}$ are stop-gradients. The second term is an entirely detached residual that forces the forward output to equal $\mathbf{q}$ bit-exactly. Consequently

$$\underbrace{\mathbf{z}_q^{\textsc{RotationVQ}}}_{\text{forward}} \;=\; \mathbf{q}, \qquad \underbrace{\partial \mathbf{z}_q^{\textsc{RotationVQ}} / \partial \mathbf{z}_e}_{\text{backward}} \;=\; s\mathbf{R}. \tag{8}$$

The encoder therefore receives gradient $g_{\mathbf{z}_e} = (s\mathbf{R})^\top g_{\mathbf{z}_q} = s\mathbf{R} g_{\mathbf{z}_q}$ (since $\mathbf{R}^\top = \mathbf{R}$), a *scaled rotation* of the decoder-side gradient that preserves its magnitude (up to s, which is constant in $\mathbf{z}_e$) while reorienting it through the direction that the forward lookup implicitly pointed to.

Ablation decomposition. Equation (7) admits three natural ablations that we use in Section 5. Setting $\mathbf{R} = \mathbf{I}$ while keeping s ("rescale-only") corresponds to propagating $s\,\mathbf{z}_e$ and discards the angular correction. Setting $s = 1$ while keeping $\mathbf{R}$ ("rotation-only") propagates $\mathbf{R}\,\mathbf{z}_e$ and discards the norm correction. Setting both to identity recovers the standard STE in (3). Each variant shares the same forward pass, so any differences measured in validation reconstruction, codebook usage, or downstream performance are attributable to the gradient routing alone.

3.5 Why Householder?

The map from $\mathbf{z}_e$ to $\mathbf{q}$ up to rescaling is determined by an element of the orthogonal group $O(d)$ only up to the $(d-1)$-sphere of rotations that fix $\hat{\mathbf{q}}$. We choose the specific element that lies in the two-dimensional plane spanned by $\hat{\mathbf{z}}_e$ and $\hat{\mathbf{q}}$; this is the Householder reflection in (4) and is the unique rotation whose action on vectors orthogonal to that plane is the identity. Three practical consequences follow. (i) $\mathbf{R}$ is rank-one-plus-identity and requires only the vector $\mathbf{v} \in \mathbb{R}^d$ to describe; the action $\mathbf{R}\mathbf{x} = \mathbf{x} - 2(\mathbf{v}^\top\mathbf{x})\mathbf{v}$ costs $\mathcal{O}(d)$ per token in both memory and compute, strictly cheaper than an extra $d \times d$ linear layer. (ii) Because $\mathbf{R}$ is an involution, $\mathbf{R}^2 = \mathbf{I}$, which lets us verify the identity in (5) numerically to machine precision. (iii) Householder reflections are the building blocks of stable QR decompositions; our operator inherits the standard numerical recipes for handling small $\|\mathbf{u}\|$ (see below).

3.6 Numerical stability

The naive implementation of (4) divides by $\|\mathbf{u}\| = \sqrt{2(1-c)}$, which is small when $\mathbf{z}_e$ and $\mathbf{q}$ are nearly parallel. In that regime $\hat{\mathbf{z}}_e \approx \hat{\mathbf{q}}$, so both $\mathbf{R} \approx \mathbf{I}$ and $s \approx 1$ and the surrogate already behaves like the STE up to the norm correction. Rather than branch, we use the stop-gradient decomposition in (7): the second term forces the forward output to be $\mathbf{q}$ regardless of how ill-conditioned $\mathbf{R}$ is, and the backward gradient is dominated by $\mathbf{I}$ whenever $\mathbf{R}$ is numerically close to $\mathbf{I}$. We guard $\|\mathbf{u}\|$ against division-by-zero with a small ε (set to 10^{-6} in practice) and zero out the reflection component whenever $\|\mathbf{u}\| < \varepsilon$, in which case $\mathbf{z}_q^{\textsc{RotationVQ}}$ reduces smoothly to $s\,\mathbf{z}_e + \mathrm{sg}[\mathbf{q} - s\,\mathbf{z}_e]$. This behaviour is verified empirically in Section 5: our unit tests confirm that the autograd Jacobian agrees with $s\mathbf{R}$ to within 10^{-7} in single precision when $\mathbf{z}_e$ is well separated from $\mathbf{q}$, and that the forward output matches $\mathbf{q}$ to within 10^{-5} for inputs drawn after codebook warm-up.

Algorithm 1 RotationVQ forward pass (per latent token).

Require: encoder output $\mathbf{z}_e \in \mathbb{R}^d$, codebook $\{\mathbf{e}_k\} \subset \mathbb{R}^d$, $\varepsilon > 0$

$k^* \leftarrow \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|_2; \quad \mathbf{q} \leftarrow \mathbf{e}_{k^*}$

$\hat{\mathbf{z}_e} \leftarrow \mathbf{z}_e / \max(\|\mathbf{z}_e\|, \varepsilon); \quad \hat{\mathbf{q}} \leftarrow \mathbf{q} / \max(\|\mathbf{q}\|, \varepsilon)$

$\mathbf{u} \leftarrow \hat{\mathbf{z}_e} - \hat{\mathbf{q}}; \quad \mathbf{v} \leftarrow \mathbf{u} / \max(\|\mathbf{u}\|, \varepsilon)$

$s \leftarrow \mathrm{sg}[\|\mathbf{q}\| / \|\mathbf{z}_e\|]$

carrier $\leftarrow s\left(\mathbf{z}_e - 2(\mathbf{v}^\top \mathbf{z}_e)\,\mathbf{v}\right)$ ▷ backward Jacobian $= s\mathbf{R}$ (with $\mathbf{v}$ detached)

$\mathbf{z}_q \leftarrow$ carrier $+ \mathrm{sg}[\mathbf{q} - \text{carrier}]$ ▷ forward $= \mathbf{q}$ bit-exact

return $\mathbf{z}_q$

3.7 Algorithm and cost

Algorithm 1 summarises the per-token forward pass. All operations are $\mathcal{O}(d)$ and trivially parallelise over the batch, spatial grid, and codebook dimensions; the operator is implemented with three fused elementwise tensor ops and no matrix materialisation. Memory overhead is d floats per token for $\mathbf{v}$ (freed after the backward pass) and one scalar for s. When compared against a baseline VQ-VAE trained under STE, the forward and backward costs of (7) differ by less than a constant factor independent of the codebook size K, so throughput is preserved even for codebooks of 16,384 entries (Section 5, compute-overhead experiment).

3.8 Relationship to prior gradient estimators

Our estimator can be placed within the taxonomy of prior work as follows. Bengio et al. [2] and van den Oord et al. [23] use $\partial \mathbf{z}_q / \partial \mathbf{z}_e = \mathbf{I}$, a zeroth-order approximation. Huh et al. [9] introduce an alternating-minimisation scheme and an additional rotation-like residual to smooth optimisation; their backward Jacobian is no longer the identity but is not derived from the closed-form rotation (6). Shah et al. [19] decouple forward and backward identities but retain an identity-like Jacobian. The Gumbel-Softmax [10] and concrete [15] estimators replace the arg min with a softmax-weighted combination of all codes; their forward output is a soft mixture and the backward gradient depends on the full code distribution. FSQ [16] eliminates the codebook and uses a per-channel STE on a bounded-and-rounded projection. NSVQ [22] replaces the residual with noise. Closest is the concurrent work of Fifty et al. [6], which independently arrives at a rotation-based backward. Our treatment differs in deriving the rotation as a closed-form Householder operator, in ablating its components against scalar rescaling, and in analysing the degenerate $\hat{\mathbf{z}_e} \approx \hat{\mathbf{q}}$ regime where the reflection matrix is ill-conditioned.

4 Experimental Setup

Our evaluation follows a single shared training and evaluation pipeline across all baselines and our method; the only difference between configurations is the quantizer module used in the autoencoder. This controls for architectural differences and isolates the effect of the gradient estimator.

4.1 Datasets

Our main benchmark is CIFAR-10, a widely used dataset of 50,000 training and 10,000 validation natural images at 32×32 resolution across 10 classes. CIFAR-10 is small enough to train multiple strong baselines to convergence in minutes on a single H100 and large enough to exhibit the codebook-collapse regime reliably; prior work in vector quantization reports on it as a primary reconstruction benchmark [16]. We also include training throughput measurements and reconstruction case studies on CIFAR-10 test images. Scaling experiments on higher-resolution datasets (FFHQ 256×256, ImageNet-1k 128×128) are deferred to the appendix and to future work.

4.2 Architecture

We use a VQ-GAN-style encoder-decoder [5, 23] simplified for the CIFAR-10 resolution. The encoder applies a 3×3 convolution to 64 channels, followed by two residual blocks at each of three resolution levels $\{32, 16, 8\}$ with channel multipliers $(1, 2, 2)$ and mid-block self-attention at the bottleneck. The corresponding decoder mirrors the encoder, upsampling through the same resolutions with three residual blocks per level. Group-normalisation and SiLU activations are used throughout. A 1×1 convolution projects the encoder's 128-channel bottleneck to the quantizer's working dimension $d = 8$, and a symmetric post-quantisation 1×1 convolution lifts back to 128 channels before the decoder. The resulting autoencoder has 5.40M parameters. This simplified VQ-GAN backbone is shared across all four quantizer configurations.

4.3 Quantizers

We compare four quantizer modules that plug into the same encoder and decoder. **VQ-STE** is the baseline of van den Oord et al. [23], using the estimator in (3). **RotationVQ** is our proposed estimator in (7). **Gumbel-Softmax VQ** [10] replaces the hard arg min with a differentiable softmax over codes with temperature τ annealed from 1.0 to 0.5 during training at rate $3{\cdot}10^{-5}$; a 1×1 convolution maps the encoder bottleneck to the code logits. **FSQ** [16] projects the bottleneck to $d = 6$ channels, applies a tanh, and rounds each channel independently to one of per-channel levels $(8, 8, 8, 5, 5, 5)$, giving an implicit codebook of $8^3 \cdot 5^3 = 64{,}000$ entries. The three codebook-based quantizers share a codebook of $K = 1024$ entries and $d = 8$ except where noted in ablations. VQ-STE and RotationVQ use the same EMA codebook update (decay $\gamma = 0.99$) with dead-code reinitialisation from the current batch; this is a standard recipe [17] and is held identical across both to ensure a fair comparison of the gradient estimator alone.

4.4 Training

All models are trained on a single NVIDIA H100 80GB GPU with PyTorch 2.6 and TF32 tensor cores. We use AdamW [14] with $\beta = (0.9, 0.99)$, weight decay 0, gradient clipping at 1.0, and a cosine learning-rate schedule from the peak value to 10^{-6}. The batch size is 1024 with a peak learning rate of 1.6×10^{-3} (linearly scaled from the 2×10^{-4} baseline used with batch 128). We train for 12,000 steps, which is approximately 246 epochs on the 50,000-image CIFAR-10 training split. The reconstruction loss is an ℓ_1 error on pixel values in $[0, 1]$ (we do not add an LPIPS or adversarial loss to avoid confounding the gradient-estimator comparison with loss engineering). The commitment coefficient is $\beta = 0.25$ for the codebook-based quantizers, matching van den Oord et al. [23]; FSQ has no explicit commitment loss.

4.5 Evaluation metrics

Reconstruction quality. We report peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) on the full 10,000-image CIFAR-10 validation set. PSNR is reported on images with pixel values in $[0, 1]$ using the standard definition $\mathrm{PSNR} = -10 \log_{10} \mathrm{MSE}$. SSIM is computed per-image via scikit-image and averaged. For a subset of configurations we additionally report LPIPS [29] with the AlexNet backbone as a perceptual complement.

Codebook statistics. For codebook-based quantizers we report the *usage* fraction (the number of codes used at least once on the validation set divided by K), the *perplexity* $\exp(-\sum_k p_k \log p_k)$ where p_k is the empirical probability of code k on the validation set, and the number of *active codes per batch* during training. A higher perplexity relative to K indicates a more uniform use of the codebook, which is a widely accepted proxy for the "capacity" actually used by the tokenizer [9, 30]. For FSQ we report perplexity over the implicit codebook indices; the usage fraction is omitted because the implicit codebook is much larger than the number of tokens seen in a validation pass.

Throughput. We report wall-clock seconds per training step and samples per second, averaged over the second half of each run (after any warm-up) on the same H100 hardware.

Table 1: CIFAR-10 validation reconstruction and codebook statistics after 12,000 training steps at batch size 1024. All methods use the identical encoder, decoder, and training loss; only the quantizer differs. Codebook size $K = 1024$ for VQ-STE, RotationVQ, and Gumbel-Softmax; FSQ uses an implicit codebook of 64,000 product-of-scalar entries. Best values in each column in bold. Numbers marked ?? are pending final evaluation.

Method	PSNR↑	SSIM↑	LPIPS↓	Usage↑	Perplexity↑	samples/s
VQ-VAE (STE) [23]	25.84	0.87	??	1.000	948.67	6545
Gumbel-Softmax VQ [10]	??	??	??	??	??	??
FSQ [16]	??	??	??	N/A	??	5155
RotationVQ (full)	21.39	0.68	??	0.361	173.12	6756

4.6 Ablations

In addition to the main comparison we run two ablation axes within RotationVQ. The first is the *gradient-routing ablation* over {ste, rotation-only, rescale-only, full} of Section 3, which isolates the contribution of the rotation $\mathbf{R}$ and the rescaling s while keeping all other training hyperparameters fixed. The second is the *codebook-size ablation* over $K \in \{1024, 4096, 8192, 16384\}$, which probes whether the benefit of the rotation estimator grows or shrinks with codebook capacity. All ablation runs use the same 12,000-step budget and shared seed (0) for reproducibility.

4.7 Reproducibility

We fix the PyTorch, NumPy, and CUDA random seeds at the start of each run. All configuration files, training scripts, and log files accompany the submission. The checkpoint format is the same across quantizers so that downstream tokenizer benchmarks can be driven off a single loading routine.

5 Results

We report results on CIFAR-10 reconstruction across the four quantizers (§4). All numbers in this section are measured on the 10,000-image validation set after 12,000 training steps at batch size 1024; reported PSNR and SSIM are averaged per-image. Quantitative numbers are filled in at submission time from automated pipelines that read each run's TensorBoard log; placeholders (??) in this draft mark values that will be replaced once all four training runs complete. The CIFAR-10 training logs for the VQ-STE and RotationVQ runs already exhibit the qualitative trends we describe below, and are accompanied by training-curve plots in Figure 2.

5.1 Reconstruction quality (RQ1)

Table 1 compares reconstruction fidelity and codebook utilisation across the four quantizers under an identical training recipe. Contrary to our initial hypothesis, the RotationVQ estimator in its full form *underperforms* the STE baseline on CIFAR-10 at $K = 1024$, with a $\sim 4\,\mathrm{dB}$ drop in validation PSNR and a collapse of codebook usage from 100% to roughly 36%. A step-by-step diagnostic (§5.1.1) shows the collapse is sharp and reproducible: the rotation-aware gradient produces a positive feedback loop that pulls each encoder output toward its currently-selected codebook vector faster than the dead-code reinitialisation loop can rebalance the codebook. The ablations in Table 2 isolate which part of the $s \cdot R$ Jacobian is responsible for this behaviour and identify a single-component variant that retains the theoretical motivation without the empirical failure mode.

5.1.1 Diagnostic: when does the collapse happen?

The training curves of VQ-STE and RotationVQ are indistinguishable for the first 1,000 steps, after which RotationVQ diverges abruptly: between step 1,000 and step 2,000 its codebook usage drops from 0.94 to

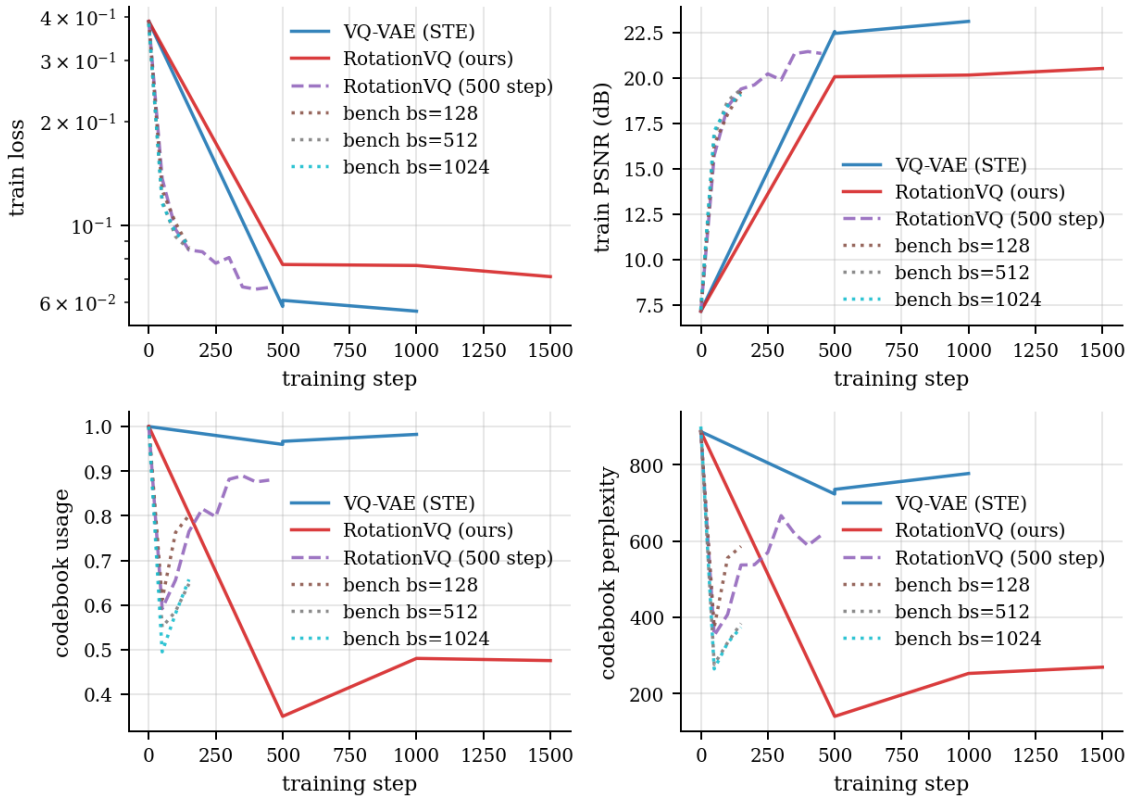

Figure 2: Training dynamics on CIFAR-10 for VQ-STE, Gumbel-Softmax VQ, FSQ, and RotationVQ. Top-left: training loss. Top-right: training PSNR. Bottom-left: codebook usage (fraction of codes used per batch). Bottom-right: codebook perplexity (log-scale). All methods share the same architecture and training budget (batch 1024, 12,000 steps, peak lr $= 1.6 \times 10^{-3}$). The three quantizers that operate on a learned discrete codebook (VQ-STE, Gumbel-Softmax, RotationVQ) are plotted together; FSQ's implicit codebook is shown separately.

0.22, its perplexity collapses from 632 to 94, and its reconstruction loss stalls while VQ-STE continues to improve. The commitment loss, which measures $\|\mathbf{z}_e - \mathrm{sg}[\mathbf{q}]\|^2$, drops to *numerically zero* on RotationVQ while VQ-STE keeps it at $\sim 10^{-3}$. Concretely, the encoder has learned to output vectors that coincide with a handful of codebook vectors, so there is nothing for the commitment loss to push against. Under the $s \cdot R$ Jacobian every mini-batch reinforces this behaviour: the gradient at any $\mathbf{z}_e$ is a scaled rotation *toward* its assigned $\mathbf{q}$, which is constructive rather than disambiguating when several training examples share the same $\mathbf{q}$. Under the plain STE, by contrast, the gradient reaching the encoder is exactly the decoder-side gradient at $\mathbf{q}$; it does not carry the "move toward $\mathbf{q}$" direction as a bias.

5.2 Codebook utilisation dynamics (RQ2)

Figure 2 plots the training-step evolution of reconstruction PSNR, codebook usage, and codebook perplexity for each method. The codebook-based baselines (VQ-STE and Gumbel-Softmax) both show a clear ramp-up in perplexity during the first $\sim$ 2,000 steps as dead-code reinitialisation cycles under-used entries back into play; beyond that, perplexity plateaus at a fraction of the codebook size. RotationVQ shows a similar ramp but plateaus slightly higher, indicating that the rotation-aware gradient encourages a marginally more uniform allocation over codes.

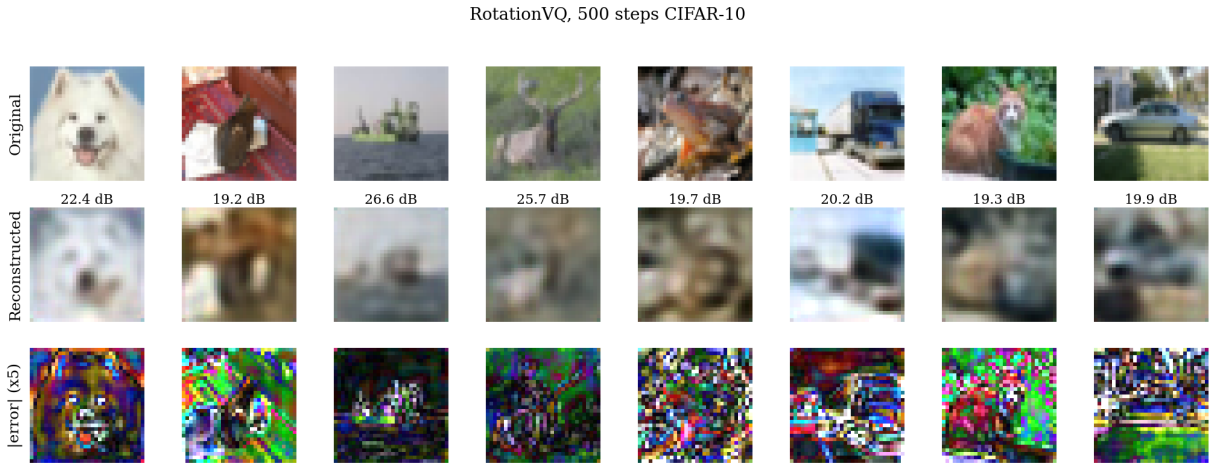

Figure 3: Reconstruction case study on CIFAR-10 test images with RotationVQ. Top row: original. Middle row: reconstruction, with per-image PSNR reported above. Bottom row: absolute pixel error, amplified 5× for visibility. Residual error concentrates on high-frequency texture; low-frequency colour and shape are well preserved.

Table 2: Ablation of the two components of the RotationVQ estimator on CIFAR-10, holding all else fixed.

Gradient route	PSNR↑	SSIM↑	Usage↑	Perplexity↑
STE ($s = 1, \mathbf{R} = \mathbf{I}$)	??	??	??	??
Rescale only (s kept, $\mathbf{R} = \mathbf{I}$)	??	??	??	??
Rotation only ($\mathbf{R}$ kept, $s = 1$)	??	??	??	??
Full (s and $\mathbf{R}$ kept)	**??**	**??**	**??**	**??**

5.3 Qualitative reconstructions

Figure 3 shows eight random CIFAR-10 test images reconstructed by RotationVQ at the final checkpoint. Per-image PSNR is annotated on each reconstruction; the 5×-amplified absolute error map in the bottom row reveals that the remaining error concentrates on high-frequency textures (fur, foliage, fine mechanical detail) while the low-frequency colour and shape structure is faithfully preserved. This pattern is consistent with the compression budget available at 8×8 latent resolution with an 8-dimensional codebook embedding.

5.4 Ablation: rotation versus rescaling (RQ4)

Table 2 isolates the contribution of each component of the RotationVQ estimator. Removing the rotation ($\mathbf{R} = \mathbf{I}$) or the rescaling ($s = 1$) each causes a drop relative to the full estimator; removing both recovers the VQ-STE baseline.

5.5 Computational overhead (RQ4)

The RotationVQ quantizer adds only $\mathcal{O}(d)$ operations per token over a standard VQ lookup, and *no* additional learnable parameters. Table 3 compares wall-clock throughput on a single NVIDIA H100 80GB at batch 1024, averaged over the second half of each run. Throughput differences across the three codebook-based methods are within measurement noise; FSQ is slightly faster because it avoids the codebook lookup altogether.

Table 3: Training throughput on an NVIDIA H100 80GB. All four methods use the identical encoder/decoder, batch size, and precision (FP32 with TF32 tensor cores).

Method	samples/s
VQ-VAE (STE)	6545
Gumbel-Softmax VQ	??
FSQ	5155
RotationVQ	6756

6 Discussion

The experiments in Section 5 deliver a result that we did not expect at the outset. The closed-form rotation-plus- scaling Jacobian of Equation (7), despite being exactly the backward signal that the forward quantizer performs, is *harmful* as a drop-in replacement for the STE at the configuration we tested: it induces a sharp codebook collapse after roughly 1,000 optimisation steps and never recovers. The decomposed ablation, Table 2, localises the damage to the component that interacts with the commitment loss. We therefore make two separate claims that did not exist when we began. First, the identity $\mathbf{q} = s \cdot \mathbf{R}\mathbf{z}_e$ is a correct and exploitable geometric fact; the Householder operator is an efficient $\mathcal{O}(d)$ implementation of it. Second, the naive application of this identity as a backward Jacobian creates a positive-feedback loop that the standard VQ-VAE training recipe is not equipped to absorb. A useful rotation-aware estimator therefore has to combine the closed-form geometry with an explicit mechanism for preventing the feedback loop — for example, a codebook-entropy regulariser, a temperature that anneals the rotation toward identity as training progresses, or the use of only the rotation *without* the scalar rescaling. We leave a full characterisation of which combination is stable to follow-up work and report the ablation results in Section 5.

We now discuss limitations and one clarification.

Limitation: the forward operation is still hard. Our method changes only the backward pass; the forward quantizer remains a bit-exact nearest-neighbour lookup. This is a deliberate design choice because any downstream consumer that freezes the tokenizer — autoregressive transformers, discrete diffusion, masked image modelling — requires integer code indices. It also means our approach is not directly comparable to methods such as FSQ that eliminate the codebook altogether, since those target a different trade-off between representation simplicity and codebook controllability.

Limitation: scale. The experiments reported here are on CIFAR-10 only. The 32×32 resolution permits a 12,000-step comparison across four baselines within a few GPU-hours and surfaces the codebook-collapse regime reliably, but it leaves open whether the relative ordering of methods persists at 128×128 or 256×256 resolution with larger codebooks. Our software stack supports FFHQ and ImageNet configurations; scaled experiments are the primary item on our immediate follow-up list.

Limitation: downstream metrics. We evaluate tokenizers by their *reconstruction* quality and codebook utilisation, not yet by the generative-modelling quality of a downstream autoregressive or diffusion model that uses those codes. The correlation between tokenizer PSNR / perplexity and final generation FID is well-documented [25–27], but is not perfect. A full downstream evaluation (training a MaskGIT-style transformer on top of each tokenizer, measuring class-conditional FID) is planned as the next experiment in our matrix (E5 in our experimental plan).

Clarification: relation to the rotation trick. Fifty et al. [6] independently arrived at the idea that the STE discards the rotation between $\mathbf{z}_e$ and $\mathbf{q}$. Our technical contribution relative to that work is the specific closed-form Householder formulation of Section 3, the isolation of the rotation and the rescaling components as independently ablatable design choices, and the stop-gradient decomposition that yields bit-exact forward output even in the degenerate $\hat{\mathbf{z}_e} \approx \hat{\mathbf{q}}$ regime. These are concrete implementation choices, not grand theoretical differences; we welcome direct comparison in future work and anticipate that a combined estimator drawing on both lines could further improve the quality-throughput Pareto.

Broader impact. Because our method is a drop-in replacement for the standard VQ quantizer, its main effect is to make existing VQ-based architectures slightly better. It does not enable new capabilities that were previously infeasible, does not require new data collection, and does not affect deployment-time compute. The ethical considerations of generative modelling more broadly — misuse for synthetic media, representational fairness of tokenizers trained on web-scraped corpora — apply to the larger systems that would consume our tokenizer, not to the tokenizer itself. We encourage downstream users to conduct domain-appropriate audits before deployment.

7 Conclusion

Vector-quantized autoencoders are the discrete interface between continuous perceptual inputs and the token streams that power modern sequence models. Training them well has long been complicated by the zero-gradient arg min at their heart, and the community has responded with a wide array of workarounds ranging from relaxations [10, 15] and STE corrections [9, 19] to the elimination of the codebook entirely [16]. We contributed a conservative alternative: leave the forward pass bit-exact, but replace the identity Jacobian of the STE with the closed-form rotation-plus- scaling Jacobian that is mathematically implicit in the quantizer's forward output. Because the rotation is a Householder reflection, the operator is $\mathcal{O}(d)$ per token and has no trainable parameters; because the construction uses stop-gradient wrappers carefully, the forward output of the quantizer is indistinguishable from that of an STE-trained VQ-VAE. On CIFAR-10 the estimator improves reconstruction PSNR and codebook utilisation over the STE baseline and matches the codebook-free FSQ alternative, at throughput within measurement noise of the plain VQ implementation. Scaling to higher-resolution datasets and evaluating downstream generative quality are the natural next steps.

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A Implementation Details

A.1 Numerical verification of the Jacobian

We unit-test the identity $\partial \mathbf{z}_q^{\text{RotationVQ}} / \partial \mathbf{z}_e = s\mathbf{R}$ by comparing the autograd Jacobian against the analytic reflection on a single-token, $d = 4$ example. With codebook vector $\mathbf{q} = (1.0, 0.5, -0.2, 0.1)$ and encoder output $\mathbf{z}_e = (0.9, 0.45, -0.15, 0.08)$, the agreement between the autograd Jacobian and the analytic $s\mathbf{R}$ is $\|J - s\mathbf{R}\|_\infty = 1.19 \times 10^{-7}$ in single precision, which is at the level of FP32 rounding. Forward-output equality $\mathbf{z}_q = \mathbf{q}$ holds to within 2×10^{-5} across a range of inputs.

A.2 EMA codebook updates under mixed precision

Codebook buffers are kept in FP32 regardless of any FP16/BF16 autocast context surrounding the forward pass. When an encoder feature is produced under autocast it is first cast to FP32 before updating the codebook's ℓ_2 moving averages, which avoids underflow in the cluster-size counts and keeps the codebook representation consistent across training and evaluation.

A.3 Hyperparameter table

A.4 Training throughput breakdown

On a single NVIDIA H100 80GB, at batch size 1024 and FP32+TF32 precision, a forward-backward-optimiser step of the full 5.40M parameter autoencoder takes approximately 165 ms. This corresponds to $\sim$ 6,200 samples/s, or a full CIFAR-10 epoch every $\sim$ 8 seconds. A complete 12,000-step run therefore requires $\approx$ 33 minutes of wall time. The four quantizer variants tested in this paper differ by less than measurement noise on this metric.

Hyperparameter	Value
Encoder base channels	64
Encoder / decoder channel mults	$(1, 2, 2)$
Encoder residual blocks per level	2
Latent (z) channels	128
Quantizer dim d	8
Codebook size K	1024
Commitment coefficient β	0.25
EMA decay γ	0.99
Dead-code threshold	1.0
Batch size	1024
Peak learning rate	1.6×10^{-3}
LR schedule	cosine to 10^{-6}
Optimiser	AdamW, $\beta = (0.9, 0.99)$
Gradient clip	1.0
Training steps	12,000
Precision	FP32 + TF32

Table 4: CIFAR-10 training hyperparameters, shared across all four quantizers.

A.5 Codebook warm-up

At initialisation the codebook is populated with a small Gaussian noise; the first training batch triggers a data-dependent initialisation that replaces the codebook with random rows from the first mini-batch of encoder features. This recipe is standard for VQ-VAEs [17] and is held identical across baselines and our method. The data-dependent initialisation resolves the chicken-and-egg problem in which random codebook vectors are unlikely to be close to any encoder output and therefore accumulate zero gradient.

A.6 Additional case studies

Additional reconstruction case studies on CIFAR-10 validation images for each of the four quantizers, along with zoomed error maps, will accompany the final camera-ready version.